```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```python
df = pd.read_csv(
    "/content/fcc-forum-pageviews.csv",
    parse_dates=["date"],
    index_col="date"
)
```

```python
df = df[
    (df["value"] >= df["value"].quantile(0.025)) &
    (df["value"] <= df["value"].quantile(0.975))
]
```

```python
def draw_line_plot():
    fig, ax = plt.subplots(figsize=(12, 6))

    ax.plot(df.index, df["value"], color="red")

    ax.set_title("Daily freeCodeCamp Forum Page Views 5/2016-12/2019")
    ax.set_xlabel("Date")
    ax.set_ylabel("Page Views")

    fig.savefig("line_plot.png")
    return fig
```

```python
def draw_bar_plot():

    df_bar = df.copy()
    df_bar["year"] = df_bar.index.year
    df_bar["month"] = df_bar.index.strftime("%B")

    df_bar = (
        df_bar
        .groupby(["year", "month"])["value"]
        .mean()
        .unstack()
    )

    month_order = [
        "January","February","March","April","May","June",
        "July","August","September","October","November","December"
    ]

    df_bar = df_bar[month_order]

    fig = df_bar.plot(kind="bar", figsize=(12,8)).figure

    plt.xlabel("Years")
    plt.ylabel("Average Page Views")
    plt.legend(title="Months")

    fig.savefig("bar_plot.png")
    return fig
```

```python
def draw_box_plot():

    df_box = df.copy()
    df_box.reset_index(inplace=True)

    df_box["year"] = df_box["date"].dt.year
    df_box["month"] = df_box["date"].dt.strftime("%b")
    df_box["month_num"] = df_box["date"].dt.month

    df_box = df_box.sort_values("month_num")

    fig, axes = plt.subplots(1, 2, figsize=(15,6))

    sns.boxplot(x="year", y="value", data=df_box, ax=axes[0])
    axes[0].set_title("Year-wise Box Plot (Trend)")
```

```
    axes[0].set_xlabel("Year")
    axes[0].set_ylabel("Page Views")

    sns.boxplot(x="month", y="value", data=df_box, ax=axes[1])
    axes[1].set_title("Month-wise Box Plot (Seasonality)")
    axes[1].set_xlabel("Month")
    axes[1].set_ylabel("Page Views")

    fig.savefig("box_plot.png")
    return fig
```

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv(
    "fcc-forum-pageviews.csv",
    parse_dates=["date"],
    index_col="date"
)

df = df[
    (df["value"] >= df["value"].quantile(0.025)) &
    (df["value"] <= df["value"].quantile(0.975))
]

def draw_line_plot():
    fig, ax = plt.subplots(figsize=(12, 6))
    ax.plot(df.index, df["value"], color="red")
    ax.set_title("Daily freeCodeCamp Forum Page Views 5/2016-12/2019")
    ax.set_xlabel("Date")
    ax.set_ylabel("Page Views")
    fig.savefig("line_plot.png")
    return fig

def draw_bar_plot():
    df_bar = df.copy()
    df_bar["year"] = df_bar.index.year
    df_bar["month"] = df_bar.index.strftime("%B")

    df_bar = (
        df_bar
        .groupby(["year", "month"])["value"]
        .mean()
        .unstack()
    )

    month_order = [
        "January","February","March","April","May","June",
        "July","August","September","October","November","December"
    ]

    df_bar = df_bar[month_order]

    fig = df_bar.plot(kind="bar", figsize=(12,8)).figure
    plt.xlabel("Years")
    plt.ylabel("Average Page Views")
    plt.legend(title="Months")
    fig.savefig("bar_plot.png")
    return fig

def draw_box_plot():
    df_box = df.copy()
    df_box.reset_index(inplace=True)

    df_box["year"] = df_box["date"].dt.year
    df_box["month"] = df_box["date"].dt.strftime("%b")
    df_box["month_num"] = df_box["date"].dt.month

    df_box = df_box.sort_values("month_num")

    fig, axes = plt.subplots(1, 2, figsize=(15,6))

    sns.boxplot(x="year", y="value", data=df_box, ax=axes[0])
    axes[0].set_title("Year-wise Box Plot (Trend)")
    axes[0].set_xlabel("Year")
    axes[0].set_ylabel("Page Views")
```

```
    axes[0].set_ylabel("Page Views")

    sns.boxplot(x="month", y="value", data=df_box, ax=axes[1])
    axes[1].set_title("Month-wise Box Plot (Seasonality)")
    axes[1].set_xlabel("Month")
    axes[1].set_ylabel("Page Views")

    fig.savefig("box_plot.png")
    return fig
```

Displaying Generated Plots

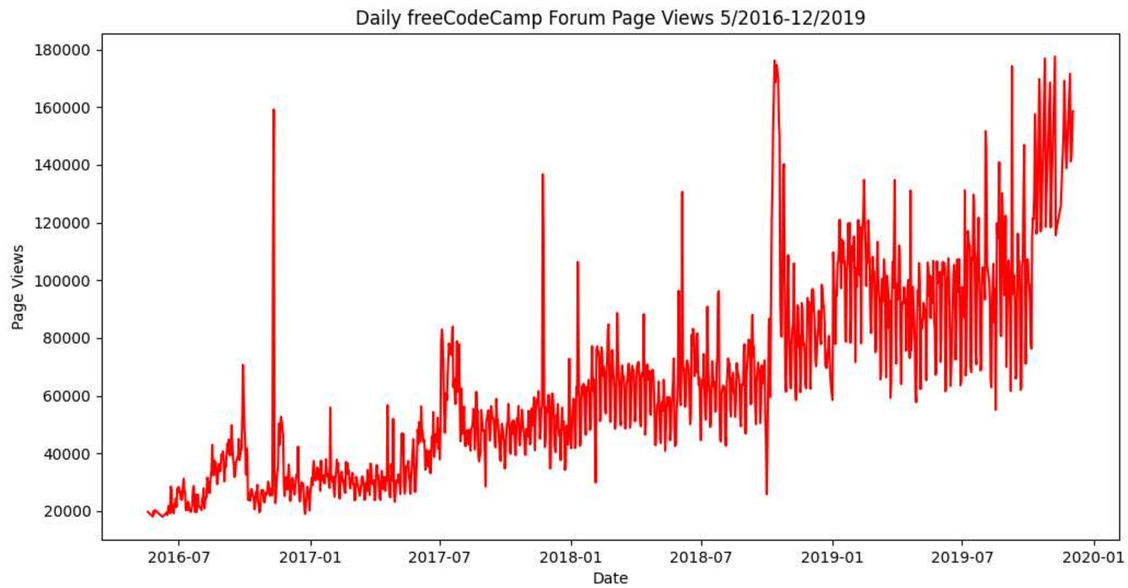
```
from IPython.display import Image

print("Line Plot:")
display(Image("line_plot.png"))

print("\nBar Plot:")
display(Image("bar_plot.png"))

print("\nBox Plot:")
display(Image("box_plot.png"))
```

```
Line Plot:
```




```
Bar Plot:
```

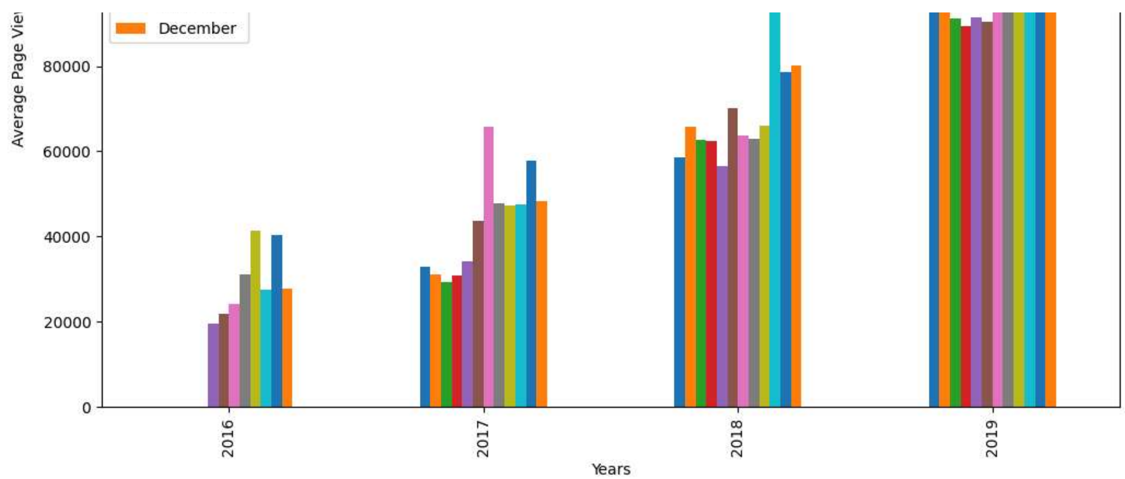
Generating Plots

```
# Generate the line plot
line_fig = draw_line_plot()

# Generate the bar plot
bar_fig = draw_bar_plot()

# Generate the box plot
box_fig = draw_box_plot()

print("Plots generated and saved as line_plot.png, bar_plot.png, and box_plot.png")
```




```
Box Plot:
```